TrajScale: Decoupled Mixed-Resolution Sampling for Efficient High-Resolution Video Generation

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Abstract

High-resolution video diffusion is computationally expensive because every denoising evaluation processes a long spatiotemporal latent sequence at the target resolution. Yet diffusion sampling exhibits a stage-wise structure: early evaluations primarily establish global layout and motion, whereas fine spatial details emerge later. This suggests executing most of the trajectory at low resolution and reserving only a short high-resolution refinement suffix. Existing low-to-high-resolution approaches, however, face distinct limitations. Conventional video super-resolution treats an already generated low-resolution video as its sole input and must independently recover the missing high-frequency content. RGB-space cascades introduce additional decoding and re-encoding overhead, while the appropriate noise level and timestep for entering a subsequent high-resolution diffusion stage remain difficult to determine. Post-generation latent upsampling avoids the codec overhead but retains this trajectory re-entry ambiguity. In-trajectory latent resizing provides a more direct alternative, yet fixed interpolation introduces substantial cross-resolution distortion, which becomes increasingly restrictive as later transitions are used to obtain greater acceleration.

We propose **TrajScale**, a decoupled mixed-resolution sampling framework built around learned clean-latent upsampling. Its **Cross-Resolution Latent Upsampler (CRLU)** learns a structurally and temporally consistent mapping between low- and target-resolution clean latent spaces, replacing fixed interpolation. However, deploying such an upsampler before the low-resolution trajectory has converged creates a timestep-dependent input mismatch: the transition-time clean prediction differs from the endpoint-like latent distribution used to train CRLU. We therefore introduce a LoRA-based **Endpoint Alignment Adapter (EAA)**, which is activated only at the transition step and calibrates the current prediction toward the endpoint of the corresponding complete low-resolution trajectory. At inference, the EAA-calibrated latent is upsampled by CRLU, reconstructed at the corresponding target-resolution schedule level, and refined by the frozen generator through a short high-resolution suffix. This division of labor lets CRLU preserve cross-scale structure while the pretrained generator synthesizes high-frequency details that are not uniquely recoverable from the low-resolution input. Both CRLU and EAA are trained independently through model-endogenous self-distillation, using supervision generated entirely by the frozen target model without external paired videos or auxiliary teachers.

On Wan2.1 with 50-step sampling and a resolution transition from 368×640 to 720×1248, TrajScale-Q and TrajScale-E achieve $1.83\times$ and $2.22\times$ end-to-end speedups. We report the six evaluated VBench dimensions separately; compared with Native-HR, both operating points preserve background consistency and motion smoothness most closely, while subject and imaging quality show larger reductions. Compared with trilinear interpolation, CRLU reduces latent L_1 error by 48.6% and LPIPS by 73.3%, while EAA reduces endpoint L_1 error by 25.8% and 21.0% at the two transition points. TrajScale also improves mixed-resolution generation with a 4-step distilled sampler, demonstrating compatibility with temporal acceleration.

1 Introduction

Recent latent video diffusion models such as Wan generate semantically coherent videos with realistic motion and fine spatial detail (Wan et al. 2025), but their inference cost grows aggressively with spatial resolution, video length, and sampling steps. In DiT-based generators, every denoising evaluation processes the full spatiotemporal token sequence. Executing the entire trajectory at the target resolution therefore forces early evaluations—which primarily establish global structure and macro-motion—to incur the same high-resolution cost as later evaluations that increasingly refine local structure and texture. Mixed-resolution inference offers a complementary form of acceleration: a low-resolution prefix models global content economically, followed by a short high-resolution suffix that completes spatial refinement. This spatial strategy (ModelTC 2026; Zheng et al. 2026) is orthogonal to temporal step distillation and can be combined with few-step samplers.

Existing routes to higher-resolution video generation leave a gap between post-generation upscaling and in-trajectory acceleration. Conventional video super-resolution receives an already completed low-resolution RGB video and must independently recover underdetermined high-frequency content (Zhou et al. 2024; Xu et al. 2024). Cascaded RGB diffusion can reintroduce a generative prior, but decoding the low-resolution result and re-encoding it for a later stage adds codec overhead, while the completed RGB video has no intrinsic correspondence to a particular timestep or noise level

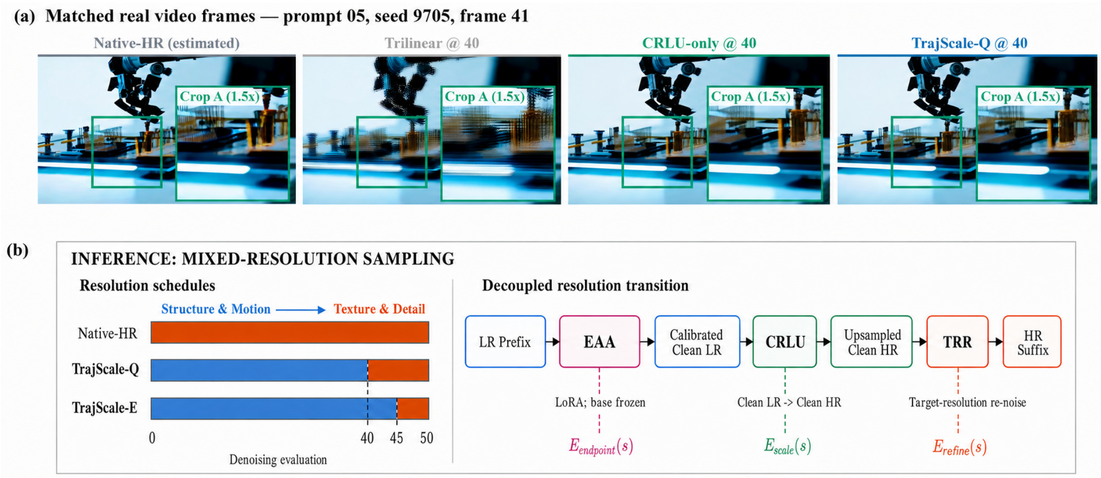

Figure 1: Real-frame comparison and TrajScale inference. (a) Content-aligned Native-HR (estimated), Trilinear, CRLU-only, and TrajScale-Q outputs for prompt 05, seed 9705, frame 41, with identical $1.5\times$ crops. The reference follows a complete 368p trajectory; transition outputs are 720p. (b) Native-HR performs all 50 evaluations at high resolution, whereas TrajScale-Q and TrajScale-E execute the first 40 and 45 evaluations at low resolution and retain 10 and 5 high-resolution suffix evaluations.

of the high-resolution trajectory. Latent cascades avoid the RGB round trip, yet still require a manually chosen trajectory entry point and refinement strength after low-resolution generation has finished. In-trajectory methods avoid this late re-entry by resizing latents during sampling, but fixed interpolation does not learn the VAE-induced correspondence between two latent grids. As the transition is delayed for greater acceleration, fewer high-resolution evaluations remain to correct interpolation error. What is missing is a learned latent upsampler that can be inserted before low-resolution convergence without inheriting the transition-time residual of an unconverged trajectory state.

We propose **TrajScale**, a decoupled mixed-resolution sampling framework organized around this requirement. TrajScale first learns a **Cross-Resolution Latent Upsampler (CRLU)** from model-generated paired clean latents, replacing geometry-only interpolation with a generator- and VAE-specific cross-resolution mapping. CRLU is designed to transfer structure and motion faithfully to the target latent grid rather than uniquely hallucinate every missing high-frequency detail; the frozen high-resolution suffix retains that generative responsibility. Deploying CRLU before the low-resolution trajectory has converged, however, presents it with a timestep-dependent input residual that is absent from its clean-latent training pairs. We therefore introduce a handoff-local **Endpoint Alignment Adapter (EAA)** that calibrates the transition-time clean prediction toward the endpoint of the corresponding native low-resolution trajectory. The aligned latent is then upsampled, reconstructed at the corresponding scheduler level, and refined by the frozen generator through the remaining high-resolution evaluations.

Our primary contributions are summarized as follows:

- **We propose TrajScale**, a lightweight mixed-resolution video sampling framework that combines an inexpensive low-resolution prefix with a short high-resolution suffix, providing controllable quality–efficiency operating points and compatibility with temporally distilled samplers.
- **We develop CRLU**, a model-endogenous clean-latent upsampler that learns the target generator's cross-resolution VAE correspondence and consistently outperforms fixed latent interpolation, while delegating underdetermined high-frequency completion to the frozen high-resolution suffix.
- **We identify and correct the deployment mismatch of learned latent upsampling** through a transition-local EAA trained with endpoint supervision and cached-prefix consistency. Experiments on standard and 4-step Wan samplers show that CRLU supplies the principal cross-resolution gain and EAA provides targeted transition-time calibration.

2 Related Work

2.1 Video Super-Resolution and Cascaded Refinement

Latent Diffusion Models move generation into a compressed representation for scalable high-resolution synthesis (Rombach et al. 2022). Conventional video super-resolution instead treats a completed low-resolution RGB video as the input to a separate restoration system (Zhou et al. 2024; Xu et al. 2024). Such a model must preserve structure and temporal consistency while also recovering high-frequency content that is not uniquely determined by the input. TrajScale does not formulate high-resolution generation as an isolated restoration problem: its learned scale operator transfers a stable latent representation, and the original generator remains responsible for underdetermined detail synthesis.

Cascaded generative systems, including latent-video pipelines and recent multi-stage systems such as LTX-2 and LUVE, improve resolution through a later upsampling or refinement stage (Blattmann et al. 2023; HaCohen et al. 2026; Zhao et al. 2026). RGB cascades incur an additional decode–encode round trip between stages and require a chosen noise level for the subsequent diffusion process. Latent cascades avoid the RGB round trip but still perform scale conversion after low-resolution completion and must determine where the upsampled latent should enter the high-resolution schedule. TrajScale instead inserts learned clean-latent upsampling before low-resolution completion and continues refinement within the frozen generator's native high-resolution schedule.

2.2 In-Trajectory Mixed-Resolution Sampling

LightX2V changes latent resolution during sampling using fixed interpolation (ModelTC 2026). RALU shows that training-free latent upscaling can suffer from high-frequency aliasing and resolution-dependent noise–timestep mismatch and develops mixed-resolution strategies for Diffusion Transformers (Jeong et al. 2025). For image flow matching, MrFlow combines low-resolution structure formation with low-noise high-resolution refinement (Zheng et al. 2026). These methods establish the efficiency potential of in-trajectory resolution changes, but fixed resizing only changes geometry; it does not learn the cross-resolution correspondence induced by a specific video VAE. TrajScale replaces interpolation with CRLU. Because a learned clean-latent upsampler is trained on converged latent pairs, applying it to an unconverged trajectory prediction introduces a distinct training–inference input mismatch, which motivates EAA.

2.3 Trajectory Consistency and Few-Step Sampling

Temporal acceleration methods reduce the number of denoising evaluations via trajectory regression, consistency mapping, or distribution matching (Luo et al. 2023; Yin et al. 2024; Zhai et al. 2024). Recent approaches such as Self-Forcing and AnyFlow further emphasize consistency between training states and actual student rollouts to reduce trajectory distribution shift (Huang et al. 2025; Gu et al. 2026). TrajScale targets spatial rather than temporal computation. EAA is not a general low-resolution step-reduction model: it is activated only at the selected transition, while the preceding prefix is generated entirely by the frozen sampler. This locality yields the cached-prefix consistency property established in Section 3.3, allowing transition states to be reused without a student-prefix rollout mismatch.

3 Method

3.1 From Clean-Latent Upsampling to In-Trajectory Transition

Let $v_\phi(x_k, k, c)$ denote a frozen Wan flow predictor, where x_k is the latent state presented to the k-th denoising evaluation, c is the text condition, and T is the total number of evaluations. Superscripts L and H denote the low- and target-resolution latent spaces. A handoff at evaluation s uses x_s^L and counts that evaluation as the final low-resolution evaluation. The frozen predictor produces an unaligned clean estimate

$$\widehat{z}_s^L = x_s^L - \sigma_s v_\phi(x_s^L, s, c),$$

where σ_s is the scheduler coefficient at the transition. We use z_{end}^L for the endpoint obtained by completing the same native low-resolution trajectory under matched prompt, seed, initial noise, scheduler, and guidance. Separately, $(z_{\text{pair}}^L, z_{\text{pair}}^H)$ denotes a pair of clean Wan-VAE latents encoded from low- and high-resolution versions of the same generated video. The distinct notation is important: the native trajectory endpoint and the paired-latent training input are both clean-like, but they arise from different conditional distributions.

The central capability required by mixed-resolution sampling is an accurate cross-resolution mapping. Fixed interpolation changes the latent grid but does not model how the frozen VAE represents the same content at two resolutions. We therefore learn a clean-latent upsampler U_ψ from paired latent examples. If U_ψ were used only after low-resolution convergence, this would be a conventional latent cascade. Acceleration instead requires deploying it at an intermediate transition, where $\widehat{z}_s^L$ still carries a step- and scheduler-dependent residual. EAA is introduced specifically to remove this transition-dependent component before CRLU is applied. Although EAA is executed before CRLU during inference, we present CRLU first because the need for endpoint alignment arises only from deploying the clean-latent upsampler before convergence.

The resulting inference path is

$$\begin{aligned} x_s^L &\xrightarrow{\text{EAA}} \widetilde{z}_s^L \xrightarrow{U_\psi} \widehat{z}_s^H \\ &\xrightarrow{\text{schedule reconstruction}} x_{s+1}^H \xrightarrow{\text{frozen HR suffix}} z_{\text{out}}^H. \end{aligned}$$

Here, $\widetilde{z}_s^L$ is the endpoint-aligned clean estimate and $\widehat{z}_s^H$ is the CRLU output on the target grid. The two learned problems are

$$\mathcal{E}_{\text{scale}} = d\big(U_\psi(z_{\text{pair}}^L), z_{\text{pair}}^H\big), \qquad \mathcal{E}_{\text{align}}(s) = d\big(\widetilde{z}_s^L, z_{\text{end}}^L\big).$$

They are not treated as additive components of a single transition error. The remaining high-resolution budget $R = T - s$ is instead an operating condition: evaluations $1, \ldots, s$ run at low resolution and evaluations $s+1, \ldots, T$ run at high resolution. A later transition reduces low-resolution cost but leaves fewer frozen high-resolution evaluations to synthesize details that the low-resolution input cannot uniquely determine.

3.2 Model-Endogenous Cross-Resolution Latent Upsampling

Why learned latent upsampling. A spatial interpolation rule can resize a tensor, but it cannot learn the VAE-induced correspondence between the source and target latent grids. CRLU is therefore trained to transfer the low-frequency structure, motion trajectory, and temporally coherent content of a clean latent into the target-resolution representation. Unlike conventional video super-resolution, its objective is not to uniquely reconstruct every missing high-frequency texture. Those degrees of freedom remain available to the frozen generator during the high-resolution suffix.

Generator-endogenous scale supervision. We use the frozen Wan model to generate high-resolution reference videos from predefined prompts and random seeds. Each video is spatially downsampled in RGB space, after which the high- and low-resolution versions are independently encoded by the same frozen Wan VAE. This produces aligned pairs $(z_{\text{pair}}^L, z_{\text{pair}}^H)$ with identical frame counts, semantics, and motion trajectories. Performing the resolution change

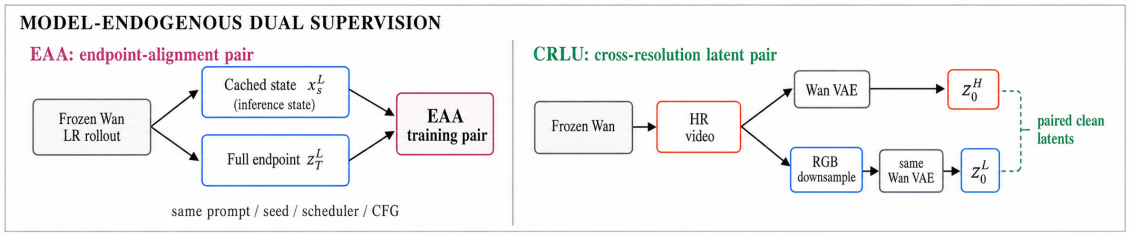

Figure 2: Overview of TrajScale. Design logic: CRLU learns a model-specific clean-latent mapping across resolutions; deploying it before low-resolution convergence creates a transition-time input residual, which EAA calibrates toward the corresponding native endpoint. Inference follows the execution order LR prefix $\rightarrow$ EAA $\rightarrow$ CRLU $\rightarrow$ scheduler-consistent state reconstruction $\rightarrow$ frozen HR suffix. Both trainable modules use supervision generated by the frozen target model.

before VAE encoding is essential: CRLU learns the actual correspondence induced by the target generator, RGB downsampling process, and VAE rather than imitating an arbitrary interpolation rule inside latent space.

Spatiotemporal latent upsampling. CRLU maps a 16-channel Wan latent to a target-resolution latent while preserving the temporal dimension:

$$U_\psi : \mathbb{R}^{B\times 16\times F\times h\times w} \rightarrow \mathbb{R}^{B\times 16\times F\times H\times W}.$$

Inspired by the latent upsampler in LTX-2 (HaCohen et al. 2026), the network follows an encoder–resampler–decoder design. A 3D convolution projects the input into a 256-dimensional hidden representation, followed by four spatiotemporal residual blocks. A spatial resampler changes only height and width; four additional 3D residual blocks and a final 3D convolution reconstruct the 16-channel output. The temporal dimension remains unchanged, while the 3D blocks allow adjacent latent frames to contribute to reconstruction.

We instantiate the spatial resampler for each latent-grid ratio. For $480 \times 832 \rightarrow 720 \times 1248$, the grid changes from 60×104 to 90×156. CRLU expands channels by a factor of nine, applies a $3\times$ spatial PixelShuffle, and performs stride-2 downsampling with a fixed binomial blur kernel:

$$60 \times 104 \xrightarrow{\times 3} 180 \times 312 \xrightarrow{/2} 90 \times 156.$$

For $368\times 640 \rightarrow 720\times 1248$, the grid changes from 46×80 to 90×156. A $2\times$ PixelShuffle produces a 92×160 grid, which is center-cropped to the target size. These resolution-specific operators implement the required rational scale changes, while the surrounding spatiotemporal network learns the latent correspondence.

Cross-resolution objectives. Given $\hat{z}^H_{\text{pair}} = U_\psi(z^L_{\text{pair}})$, CRLU is trained with

$$\mathcal{L}_{\text{CRLU}} = \mathcal{L}_{\text{rec}} + \lambda_{\text{content}}\mathcal{L}_{\text{content}} + \lambda_{\text{temp}}\mathcal{L}_{\text{temp}},$$

where

$$\mathcal{L}_{\text{rec}} = \mathbb{E}\left[\sqrt{(\hat{z}^H_{\text{pair}} - z^H_{\text{pair}})^2 + \epsilon^2}\right],$$

and

$$\mathcal{L}_{\text{content}} = \|D(\hat{z}^H_{\text{pair}}) - z^L_{\text{pair}}\|_1, \tag{1}$$

$$\mathcal{L}_{\text{temp}} = \|\Delta_F \hat{z}^H_{\text{pair}} - \Delta_F z^H_{\text{pair}}\|_1. \tag{2}$$

Here, D maps the prediction back to the low-resolution grid, and Δ_F denotes first-order temporal differences between adjacent latent frames. We use $\lambda_{\text{content}} = 0.2$, $\lambda_{\text{temp}} = 0.1$, and $\epsilon = 10^{-3}$. The three terms constrain target-latent reconstruction, low-frequency content preservation, and temporal evolution, respectively.

A low-resolution latent does not uniquely specify every texture in its high-resolution counterpart. CRLU therefore produces a structurally faithful and temporally stable clean latent on the target grid, while the frozen high-resolution suffix supplies the generative degrees of freedom absent from the input. This division distinguishes TrajScale from a standalone super-resolution model that must recover all missing detail by itself.

3.3 Transition-Time Endpoint Alignment

Deployment mismatch. CRLU is trained on paired clean latents, but mixed-resolution acceleration presents it with the clean prediction of an unconverged trajectory state. In general,

$$p^{(s)}_{\text{handoff}}(\hat{z}^L_s) \neq p_{\text{pair}}(z^L_{\text{pair}}).$$

The discrepancy contains a transition-specific residual whose magnitude and structure depend on the handoff step, scheduler, and guidance configuration. Passing this residual directly through CRLU entangles scale conversion with unfinished low-resolution denoising. EAA is therefore trained for CRLU deployment rather than as a general-purpose low-resolution acceleration model: it moves the transition-time prediction toward an endpoint-like clean-latent domain before scale conversion. It reduces, but does not claim to eliminate, the remaining gap between native trajectory endpoints and CRLU's paired-latent training distribution.

Handoff-local endpoint supervision. For each training condition, we run the frozen model with a fixed text prompt, random seed, initial noise, scheduler, and Classifier-Free

Guidance (CFG) configuration. We cache x_s^L immediately before the designated transition and continue the same native low-resolution trajectory to obtain z_{end}^L. The pair $(x_s^L, z_{\text{end}}^L)$ preserves the exact semantic and stochastic conditions of the handoff while isolating the residual that would otherwise be passed to CRLU.

Localized low-rank adaptation. EAA is implemented with LoRA (Hu et al. 2022) and is activated only at the designated transition step. The base Wan parameters remain frozen. For a linear layer W, the training-time LoRA parameterization is

$$W' = W + \frac{\alpha_{\text{LoRA}}}{r} BA,$$

where r is the LoRA rank and α_{LoRA} is its training scale. At inference, we separately modulate the learned branch by a handoff strength λ:

$$W_\lambda = W + \lambda \frac{\alpha_{\text{LoRA}}}{r} BA.$$

We sweep $\lambda \in \{0.5, 0.75, 1.0\}$ on the fixed development prompts, select $\lambda = 0.75$, and use this single value for every prompt and all reported handoff configurations. This inference-time strength is distinct from the training setting $\alpha_{\text{LoRA}}/r = 32/32$. It is nonzero only during evaluation s and is reset to zero for every other evaluation. Low-rank updates are injected into the q, k, v, and o attention projections and the ffn.0 and ffn.2 linear layers. Under the flow-matching parameterization, EAA produces

$$\tilde{z}_s^L = x_s^L - \sigma_s v_{\phi+\lambda\Delta\theta_s}(x_s^L, s, c).$$

We optimize

$$\mathcal{L}_{\text{EAA}} = \|\tilde{z}_s^L - z_{\text{end}}^L\|_1 + 0.1\|\tilde{z}_s^L - z_{\text{end}}^L\|_2^2.$$

For earlier transitions, where frame-wise residuals are more likely to propagate through scale conversion, we additionally apply a temporal residual alignment term with weight 0.05. Although the same target can be expressed as a velocity label $v^* = (x_s^L - z_{\text{end}}^L)/\sigma_s$, all reported models are optimized in the clean-latent domain. EAA performs no spatial upsampling; its sole role is to calibrate the input that CRLU receives at the selected transition.

Cached-prefix consistency. Because EAA is inactive before step s, its low-rank parameters satisfy $\Delta\theta_k = 0$ for all $k < s$. Under deterministic sampling and matched text embeddings, initial noise, scheduler, CFG scale, and other conditioning inputs, the adapted and frozen processes therefore use identical transition functions throughout the prefix. By induction,

$$x_k^{\text{adapted}} = x_k^{\text{base}}, \qquad \forall k \le s.$$

Consequently, a prefix state cached from the frozen model is exactly the state consumed by EAA. This property supports offline pair construction and controlled comparisons among transition operators without introducing prefix rollout mismatch. The equality requires transition-local activation, deterministic prefix sampling, and matched schedulers, guidance, and conditioning. Extending adaptation to earlier or multiple evaluations would alter subsequent states and require on-policy student rollouts or synchronized teacher trajectories.

3.4 Scheduler-Consistent High-Resolution Continuation

After endpoint alignment and CRLU, TrajScale has a target-grid clean latent

$$\hat{z}_s^H = U_\psi(\tilde{z}_s^L),$$

but the next denoising evaluation expects an intermediate state at the corresponding scheduler level. We therefore reconstruct

$$x_{s+1}^H = (1 - \sigma_{s+1})\hat{z}_s^H + \sigma_{s+1}\epsilon^H, \qquad \epsilon^H \sim \mathcal{N}(0, I).$$

The noise is sampled directly on the target-resolution grid. This parameter-free operation completes the scale transition in the clean-latent domain and then initializes the target-grid state at the intended scheduler coefficient, rather than resizing a low-resolution noisy state or velocity field across grids.

Starting from x_{s+1}^H, the unmodified Wan denoiser performs the remaining evaluations in the high-resolution latent space. The suffix refines the transferred structure and uses the base model's native prior to synthesize fine textures and spatial frequencies that are not recoverable from the low-resolution latent alone. State reconstruction merely enables this native continuation; the frozen high-resolution suffix, not the reconstruction operation itself, performs generative refinement.

The same procedure applies to conventional many-step and temporally distilled few-step samplers. For any valid transition index s, the sampler executes its prefix at low resolution, invokes EAA and CRLU once, reconstructs the next scheduler state on the target grid, and completes the remaining evaluations at high resolution. TrajScale neither changes the base scheduler nor retrains the frozen denoiser. Concrete transition indices and quality- or efficiency-oriented operating points are specified in Section 4.

4 Experiments

4.1 Experimental Objectives and Setup

Our evaluation addresses four questions: (RQ1) whether TrajScale establishes a useful end-to-end quality–efficiency frontier; (RQ2) whether CRLU learns a more faithful cross-resolution operator than fixed interpolation; (RQ3) whether EAA makes CRLU more reliable when deployed on an unconverged low-resolution trajectory; and (RQ4) whether the same spatial transition remains compatible with a 4-step distilled sampler.

For the 50-step pipeline, we generate approximately 1,000 reference videos using Wan2.1 T2V-1.3B. Each video contains 81 frames at 720 × 1248 resolution and 16 fps. Bicubic RGB downsampling produces corresponding 480×832 and 368×640 videos, which are re-encoded by the frozen Wan VAE to construct aligned cross-resolution latent pairs. EAA training pairs consist of the low-resolution state immediately preceding a designated transition and the endpoint of the corresponding complete low-resolution trajectory.

The 4-step experiments use approximately 5,000 videos generated by Wan2.1 T2V-14B StepDistill-CfgDistill. CRLU is trained on $368 \times 640 \rightarrow 720 \times 1248$ clean latent pairs, while EAA pairs the state preceding step 3 with the complete step-4 low-resolution endpoint. Training, validation, and test partitions are separated by both text prompts and random seeds.

CRLU is optimized with AdamW for up to 50k steps using a learning rate of 1×10^{-4}, $bf16$ precision, and an EMA decay of 0.9999. EAA is trained for up to 10k steps with AdamW, a learning rate of 5×10^{-5}, and LoRA rank/scale $32/32$. Training uses four NVIDIA H100 GPUs and requires approximately 8 hours for CRLU and 33 hours for EAA. All efficiency measurements are averaged over five independent cold-start runs. Since all supervision is generated by the frozen target model, TrajScale requires neither external paired videos nor an additional upscaling teacher.

4.2 Baselines and Evaluation Protocol

System-level comparisons include **Native-HR Sampling**, which executes all denoising evaluations at 720p; **Trilinear Handoff**, which performs mixed-resolution inference using fixed trilinear latent interpolation; and **Post-Generation Latent Cascade**, which completes the low-resolution trajectory, applies CRLU, and performs one manually selected low-intensity high-resolution refinement evaluation. We additionally report **Native-LR Sampling** as a low-resolution reference. The cascade baseline tests the alternative of postponing scale conversion until generation has finished, when the high-resolution trajectory entry level must be chosen externally.

Module interactions are evaluated through a 2×2 diagnostic design that crosses CRLU input states $\{\text{Unaligned}, \text{EAA-aligned}\}$ with resolution operators $\{\text{Trilinear}, \text{CRLU}\}$. The four configurations are Trilinear Handoff, EAA+Trilinear, CRLU-only, and the complete TrajScale framework. The comparison from Trilinear to CRLU-only isolates the learned upsampling effect; the comparison from CRLU-only to TrajScale isolates transition-time input alignment. EAA+Trilinear is retained as a diagnostic to show that endpoint calibration cannot replace an accurate scale operator. Oracle input-domain variants and a joint transition mapper are reported only as diagnostic references in the Supplementary Material.

For endpoint alignment, we report endpoint latent L_1, decoded PSNR and LPIPS, temporal latent differences, and sample-wise win rates. CRLU is evaluated using latent L_1, PSNR, SSIM, LPIPS, and temporal L_1. End-to-end quality is measured with VBench (Huang et al. 2024) and human evaluation. Rather than forming a custom aggregate, we report all six evaluated VBench dimensions directly: subject consistency, background consistency, motion smoothness, dynamic degree, aesthetic quality, and imaging quality.

All paired configurations share identical prompts, random seeds, initial noise, schedulers, and guidance parameters. Prompt-level score differences are analyzed using 10,000 paired bootstrap iterations for 95% confidence intervals and two-sided exact sign tests. Human evaluation uses randomized, double-blind comparisons over 10 matched prompts, with three independent judges per prompt and prompt-level majority voting. Inter-rater agreement is reported in the Supplementary Material.

Efficiency measurements include end-to-end cold-start latency, peak GPU memory, low- and high-resolution denoising evaluations, and speedup over Native-HR Sampling. All EAA, CRLU, VAE, scheduler-state reconstruction, and dynamic model-loading overheads are included.

5 Results and Analysis

5.1 System-Level Quality–Efficiency Trade-off

Table 1 compares the standard 50-step Native-HR pipeline with the two principal TrajScale operating points. TrajScale-Q moves the first 40 evaluations to the low-resolution grid and retains 10 high-resolution suffix evaluations, reducing latency from 253.10 to 138.36 seconds ($1.83\times$). TrajScale-E retains only five high-resolution evaluations and further reduces latency to 114.26 seconds ($2.22\times$). Across the six reported VBench dimensions, the two operating points remain closest to Native-HR in background consistency and motion smoothness, while subject and imaging quality show larger reductions. Dynamic degree increases from 0.60 to 0.70 for both variants. These operating points show that later transitions increase acceleration while reducing the high-resolution budget available for native detail synthesis.

The Post-Generation Latent Cascade reaches 86.45 seconds ($2.93\times$); relative to Native-HR, its subject, background, aesthetic, and imaging scores decrease to 0.9259, 0.9545, 0.5710, and 0.5732, respectively. Completing the low-resolution trajectory before scale conversion maximizes speed, but leaves only a manually selected low-noise refinement evaluation and therefore less capacity for structural and texture refinement. Peak memory remains approximately 26 GiB across Native-HR, TrajScale-Q, TrajScale-E, and the post-generation cascade, indicating that TrajScale primarily reduces runtime rather than peak memory.

5.2 Component Analysis

Cross-resolution latent upsampling. Table 2 isolates the cross-resolution operator before considering transition-time mismatch. Over 50 paired clean latent samples, CRLU substantially outperforms trilinear interpolation. For $480\text{p} \rightarrow 720\text{p}$, it reduces latent L_1 from 0.1929 to 0.0991, a 48.6% reduction; improves decoded PSNR by 5.71 dB; reduces LPIPS from 0.2358 to 0.0629; and lowers temporal L_1 by 30.8%. CRLU wins on all 50 samples for PSNR, LPIPS, and temporal consistency, and on 49/50 samples for SSIM.

The more aggressive $368\text{p} \rightarrow 720\text{p}$ transition shows the same trend: latent L_1, LPIPS, and temporal L_1 decrease by 33.4%, 51.1%, and 15.5%, respectively. These improvements show that the primary cross-resolution capability comes from learning the VAE-specific latent correspondence rather than relying on a geometry-only resize rule.

Transition-time endpoint alignment. With the scale operator fixed, EAA tests whether CRLU can be deployed more reliably on an unconverged trajectory prediction. At inference strength $\lambda = 0.75$, EAA reduces residual low-resolution endpoint L_1 by 25.82% at step 40 and 21.03% at step 45. Both settings improve all 10 evaluation samples, with paired confidence intervals excluding zero. EAA also increases decoded

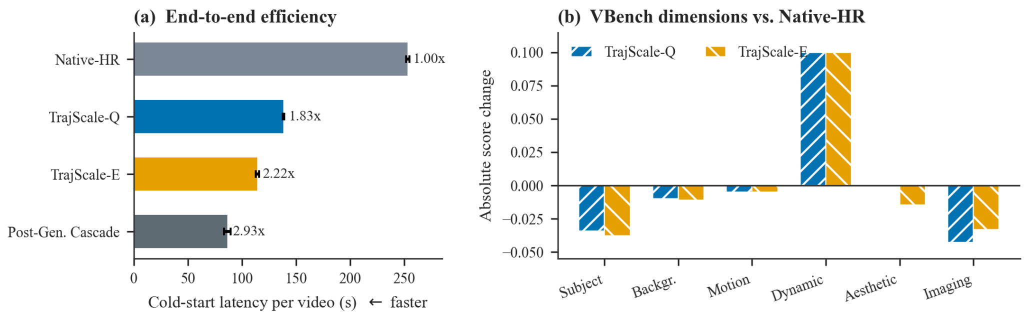

Figure 3: System efficiency and per-dimension quality relative to Native-HR Sampling. (a) End-to-end cold-start latency and speedup. (b) Absolute changes in the six evaluated VBench dimensions; dynamic degree is reported as a motion characteristic rather than treated as a monotonic quality score.

(a) End-to-end efficiency

Method	LR/HR evals.	Time (s) $\downarrow$	Latency red.	Speedup $\uparrow$
Native-HR Sampling	0/50	253.10 ± 1.53	—	$1.00\times$
TrajScale-Q (Step 40)	40/10	138.36 ± 0.50	45.33%	$1.83\times$
TrajScale-E (Step 45)	45/5	$\mathbf{114.26 \pm 1.52}$	**54.85%**	$\mathbf{2.22}\times$
Post-Generation Latent Cascade	50/1	86.45 ± 2.87	65.84%	$2.93\times$

(b) Original VBench dimensions

Method	Subject	Background	Motion	Dynamic	Aesthetic	Imaging
Native-HR Sampling	0.9614	0.9696	0.9848	0.6000	0.5949	0.6311
TrajScale-Q (Step 40)	0.9271	0.9595	0.9798	0.7000	0.5945	0.5883
TrajScale-E (Step 45)	0.9236	0.9584	0.9796	0.7000	0.5803	0.5977
Post-Generation Latent Cascade	0.9259	0.9545	0.9800	0.7000	0.5710	0.5732

Table 1: System-level comparison. Timings average five cold-start runs; the six VBench dimensions are evaluated under the custom-input protocol on ten matched prompts. All processing overheads are included.

Input $\to$ Output	Operator	Latent $L_1 \downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Temp. $L_1 \downarrow$
480p $\to$ 720p	Trilinear	0.1929	24.29	0.7159	0.2358	0.02121
480p $\to$ 720p	CRLU	**0.0991**	**30.00**	**0.8696**	**0.0629**	**0.01468**
368p $\to$ 720p	Trilinear	0.2444	23.42	0.6740	0.3352	0.02388
368p $\to$ 720p	CRLU	**0.1628**	**24.10**	**0.7592**	**0.1640**	**0.02017**

Table 2: CRLU versus trilinear interpolation on paired clean latents ($n = 50$).

PSNR by 2.55 and 2.23 dB and reduces latent temporal-difference errors by 23.44% and 21.24%, respectively. On the 4-step distilled model, it yields a smaller but consistent 5.03% endpoint-error reduction. All three comparisons have a 10/0 paired win rate and two-sided exact sign-test $p = 0.00195$; full confidence intervals are reported in Supplementary Table S2.

Inference-strength validation is non-monotonic. The intermediate value $\lambda = 0.75$ produces the lowest endpoint error and strongest downstream quality, whereas weaker or full-strength correction is less effective. EAA should therefore be interpreted as calibrated transition adaptation rather than an unconstrained replacement for the frozen predictor.

Hierarchical module effects. The 2×2 diagnostic in Table 3 confirms the hierarchical roles of the two learned modules. Replacing trilinear interpolation with CRLU provides the principal broad quality gain, with the largest changes in aesthetic and imaging quality across the three pipelines. With CRLU fixed, EAA has a narrower transition-specific effect: it improves the imaging-quality dimension by $+0.00599$, $+0.00254$, and $+0.00479$ at step 40, step 45, and on the 4-step sampler, respectively, while its changes in the other dimensions are smaller and not uniformly positive.

The comparison supports the proposed causal chain. CRLU establishes the cross-resolution representation and supplies the dominant improvements in motion smoothness, aesthetics, and overall imaging quality. EAA does not replace this capability; it calibrates the transition-time input that CRLU receives, enriching localized textures and fine structures that may be underrepresented by aggregate automatic metrics. The remaining frozen high-resolution suffix then performs native generative refinement.

5.3 Perceptual Effects and Distilled-Sampler Compatibility

Human evaluation confirms that EAA's transition-specific correction is perceptually visible on top of the same CRLU operator. EAA receives fine-detail win/loss/tie counts of $8/2/0$ at step 40 and $9/0/1$ at step 45. Overall-quality judgments are also positive, reaching $6/2/2$ and $6/0/4$, re-

Sampler / transition	Marginal effect	Subject	Background	Motion	Dynamic	Aesthetic	Imaging
Wan50 / Step 40	CRLU	+.00403	-.00114	+.00418	+.10000	+.04595	+.10120
	EAA | CRLU	+.00131	+.00145	+.00015	.00000	-.00455	+.00599
Wan50 / Step 45	CRLU	+.00228	-.00160	+.00429	.00000	+.05743	+.14089
	EAA | CRLU	-.00132	+.00019	-.00004	.00000	-.00386	+.00254
Distill4 / Step 3-of-4	CRLU	+.00344	+.00275	+.00453	.00000	+.06140	+.12156
	EAA | CRLU	-.00050	-.00186	-.00014	.00000	-.00180	+.00479

Table 3: Per-dimension VBench effects in the end-to-end factorial ($n = 10$). "CRLU" denotes CRLU-only minus Trilinear; "EAA | CRLU" denotes TrajScale minus CRLU-only. Dynamic degree is reported without assuming that a larger value always means higher quality.

Comparison	Details	Overall	Temporal
EAA vs. Unaligned, CRLU (S40)	8/2/0	6/2/2	1/9/0
EAA vs. Unaligned, CRLU (S45)	**9/0/1**	**6/0/4**	0/8/2
CRLU-only vs. Trilinear (S45)	**10/0/0**	**10/0/0**	**8/0/2**
TrajScale-E vs. Trilinear (S45)	**10/0/0**	**10/0/0**	**9/0/1**

Table 4: Prompt-level majority votes (W/L/T) over ten prompts with three raters per prompt. Bold entries have exact sign-test $p < 0.05$.

spectively. These preferences support the dimension-level VBench observation that endpoint alignment selectively improves imaging quality and localized structure.

The additional fine-scale content introduces a spatial–temporal trade-off: temporal stability favors the unaligned CRLU state in both settings. This is consistent with EAA acting as a deployment adapter rather than a uniformly beneficial denoising module: it shifts the operating point toward sharper local rendering, while CRLU and the remaining high-resolution suffix provide the broader spatial and temporal reconstruction.

CRLU provides the broader perceptual improvement. At step 45, both CRLU-only and TrajScale-E obtain $10/10$ prompt-level wins over trilinear handoff for fine details and overall quality, while also improving temporal stability. EAA adds a narrower local-detail refinement on top of this learned cross-resolution reconstruction.

TrajScale is also compatible with the 4-step distilled Wan sampler. On the 368p trajectory, TrajScale-D4 improves over Trilinear Handoff in subject consistency (0.96226 vs. 0.95932), background consistency (0.97026 vs. 0.96937), motion smoothness (0.98815 vs. 0.98376), aesthetic quality (0.64914 vs. 0.58954), and imaging quality (0.71418 vs. 0.58783), while both have dynamic degree 0.50. Cold-start latency increases only from 167.09 to 170.29 seconds. As in the 50-step setting, CRLU supplies the primary cross-resolution quality improvement, while EAA reduces the transition-time endpoint residual and improves the imaging-quality component. This result supports the orthogonality of spatial mixed-resolution sampling and temporal step distillation.

Qualitative comparisons in Figure 1(a) further illustrate CRLU's recovery of high-frequency textures and EAA's localized adjustments. Extended spatial crops and temporal sequences are provided in Figure S1. The content-matched `Native-HR (estimated)` reference is produced from a complete 368p trajectory and is used only for frame-level structural comparison; true 720p Native-HR Sampling is used for the quantitative evaluation in Table 1.

6 Discussion and Limitations

TrajScale separates cross-resolution representation learning from transition-specific input calibration. CRLU transfers structure and motion to the target latent grid, EAA makes this learned mapping usable before low-resolution convergence, and the frozen high-resolution suffix completes underdetermined detail synthesis. Scheduler-consistent state reconstruction is a parameter-free bridge between the clean-latent mapping and the native suffix; it enables, rather than independently performs, high-resolution refinement. This hierarchy matches the empirical results: CRLU provides the broad quality gain, while endpoint metrics, imaging-quality changes, and human preferences show that EAA corrects a narrower transition-time failure mode.

Both learned modules use model-endogenous supervision. The frozen Wan generator supplies high-resolution videos for paired CRLU latents and complete low-resolution trajectories for EAA, avoiding external paired datasets, standalone super-resolution teachers, or modification of the base denoiser. The two supervision streams are nevertheless not identical distributions. EAA reduces the residual between an intermediate prediction and its native low-resolution endpoint, but does not eliminate the remaining gap between native endpoints and the RGB-downsampled paired latents used to train CRLU.

Several limitations remain. CRLU is tied to the target VAE, generator distribution, resolution ratio, and downsampling process, so cross-model and cross-degradation transfer remain unverified. Fixed handoff steps require step-specific EAA parameters; timestep-conditioned adapters would be needed for adaptive transitions. Quality also depends on the manually selected suffix length and on whether the scheduler-state reconstruction remains appropriate for other solver families. Finally, TrajScale addresses model-generated in-trajectory spatial scaling rather than real-world degradations such as compression, blur, or sensor noise, and inherits the base generator's content risks and safety requirements.

7 Conclusion

We presented TrajScale, a learned in-trajectory latent upsampling framework for efficient high-resolution video generation. CRLU establishes a model-specific cross-resolution clean-latent mapping, EAA corrects the input mismatch created when that mapping is deployed before low-resolution convergence, and the frozen high-resolution suffix synthesizes details that are underdetermined by the low-resolution input. Both learned modules are trained from supervision generated by the frozen target model, and the base denoiser and scheduler remain unchanged.

TrajScale-Q and TrajScale-E achieve $1.83\times$ and $2.22\times$ speedups. Their six separately reported VBench dimensions show the smallest reductions from Native-HR in background consistency and motion smoothness and larger reductions in subject and imaging quality. CRLU consistently outperforms trilinear interpolation and supplies the primary cross-resolution quality gain; EAA reduces endpoint residuals and improves localized visual detail across standard and distilled samplers. These results show that accurate learned scale transfer, transition-specific input calibration, and native suffix refinement provide a practical division of labor for mixed-resolution video diffusion.

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